

2026

2026 GIMC
github.com/xmb505/Haku_EET

MI / WinCC	
	PLC vs
	PLC WiFi
	IP

ET	

	AI



	WinCC WinCC 6

HMI / WinCC

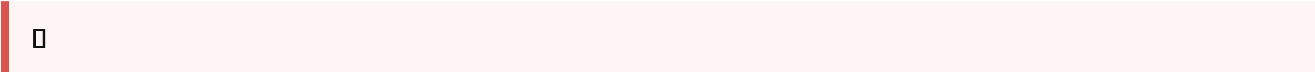
2.1

HMI WinCC

```
graph TB
  subgraph Win32["Win32"]
    subgraph WV2["WebView2"]
      WEB["Web HMI  
HTML / CSS / JS"]
    end
  end
end
```

- Win32 WebView2
- Web HMI WebView2
- WinCC

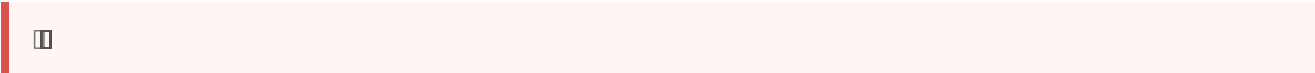
2.2



PP



PLC	SCL	
	LC IO	



3.1 PLC



PLC	
PLC	PLC
HMI	
AI	AI AI

△ MCP AI PLC

3.2

PLC	
	IP/PLC

4.1 PLC

PLC	
S7-1215C AC/DC/RLY	

PLC

4.2 WiFi

PLC WiFi 802.11

```
graph TB
    subgraph PLC["S7-1215C PLC"]
        PORT1["1"]
        PORT2["2"]
    end

    subgraph WIFI["WiFi "]
        ETH["802.11"]
    end

    LAPTOP[" "]
    EET[" "]

    PORT1 -->| " | | " ETH
    ETH -->| "802.11 | | " LAPTOP
    PORT2 -->| " | | " EET
```

- WiFi PLC
 -
 -
- S7-1215C — WiFi

4.3

⚠

	77

⚠

5.1 IP

①	IP	
②		PLC
③	EET IP	EET IP

5.2

EEP IP

IP EEP

5.3

△

EEP

- GPU EEP
-
- IO tick

EEP

6.1

PC211

6.2

VMware EEP

	3 EEP
EEP	
	2026
	EEP

6.3

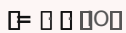
△

- EEP
-
-

6.4 □



OpenWrt	OpenWrt OpenClash
	PLC



[SPEC.md](#) [HANDOVER.md](#)

7.1

```
graph TB
    subgraph IO [IO]
        APP["app.py - IO"]
        ALGO["algorithm.py - IO"]
        PM["passenger.py - IO"]
        CONSOLE["console.py - REPL IO"]
    end

    subgraph EXEC [EXEC]
        EXEC["executor.py - FSM"]
        CTRL["controllers.py - IO"]
        UI["ui.py - IO"]
        DISP["display.py - IO"]
    end

    subgraph IO_MAP [IO MAP]
        IO["io_client.py - WS bitmap + HTTP IO"]
        MAPPER["io_mapper.py - DB+I IO"]
        VPLC["virtual_plc.py - PLC"]
    end

    APP --> ALGO
    APP --> PM
    APP --> EXEC
    EXEC --> CTRL
    EXEC --> UI
    EXEC --> DISP
    CTRL --> IO
    UI --> IO
    DISP --> IO
    IO --> MAPPER
```

□

□	□	□
□	□	Car IO □
□	□	Car ↔ IO
□	WS bitmap HTTP POST □	□
cron□	□	EventRule reschedule/cancel
watchdog□	CLOSING □	executor wait_done □

7.2 □

```
graph LR
    PLC["PLC □"] -->|"WS bitmap"| IO["□ io_client"]
    IO -->|"IO □"| APP["□ app.py"]
    APP -->|"□"| ALGO["algorithm"]
    ALGO -->|"Action"| QUEUE["ActionQueue"]
    QUEUE -->|"□"| EXEC["□ executor"]
    EXEC -->|"IO □"| IO
    IO -->|"HTTP POST"| PLC
```

```
[ sleep() [ while: poll □—□
```

7.3 □

```
graph TB
    subgraph Python_asyncio [Python asyncio]
        CORE["core □"]
    end

    subgraph I2HTTP [I2HTTP □]
        HTTP["HTTP :8080 – □DB □"]
        WS["WebSocket :8081 – □I □"]
    end

    subgraph PLC_HARDWARE ["S7-1215C PLC"]
        DB["DB □"]
        INPUT["I □"]
    end

    CORE -->|"HTTP POST<br/>tick flush 50Hz"| HTTP
    WS -->|"bitmap 800"| CORE
    HTTP --> DB
    INPUT --> WS
```

tick .io.set(addr, val) □ HTTP POST □ buffer 20ms tick □ HTTP POST □ S7 read-modify-write □ PLC □

7.4

Car	
	IO
sleep/	
	io_config.yaml gen_io_config.py
IO	tick flush
cron	
	cron watchdog CLOSING

7.5

executor action loop

CLOSING is door_close_done wait_done

CLOSING

core/watchdog.py

8.1

```
# ① I2HTTP Haku_EET
cd ~/GPIOSERVER/I2HTTP && python3 daemon_gpio.py
```

```
# ② Haku_EET
python3 -m core --simulate #
python3 -m core # I2HTTP
```

8.2

#	Command	Response
1	<code>bitmap</code>	
2		<code>/car all init down</code>
3	<code>init</code>	<code>/car 1 status</code> → <code>ready</code>
4		<code>/usermode true</code>
5		
6		<code>/usermode false</code>
7		<code>/quit</code>

8.3

Command	Response	Action
<code>IO2HTTP</code>	<code>config.yaml</code> → <code>io2http</code>	<code>/reload</code>
<code>tick</code>	<code>config.yaml</code> → <code>tick_interval_ms</code> [20ms = 50Hz]	<code>/reload</code>
	<code>config.yaml</code> → <code>per_car_init</code>	<code>/reload</code>
	<code>config.yaml</code> → <code>competition_init_timeout</code> [240s]	<code>/reload</code>
	<code>config.yaml</code> → <code>door_watchdog_timeout</code> [5s]	<code>/reload</code>
	<code>config.yaml</code> → <code>per_car_weight</code>	<code>/reload</code>
	REPL <code>/settings slow_brake <0-6></code>	
	<code>config.yaml</code> → <code>car_ids</code>	

8.4 Checklist

#	Command	Response	Action
1		<code>0</code> → <code>1</code> →	<code>0</code> → <code>1</code> →
2		<code>0</code> →	<code>0</code> → <code>0</code> →
3		<code>0</code> →	<code>0</code> → <code>0</code> →
4		<code>done=1</code> →	<code>/door 1 open</code> → <code>door_open_done</code>
5		<code>0</code> →	<code>0</code> → level ↑ ↓
6		<code>1</code> →	<code>0</code> →

`/debug show input_change` → `/car 1 manual` →

[QUICKREF.md](#) [COMMAND_MANUAL.md](#)



SPEC.md §10

P0 —

	IO tick GET
	IP
Bitmap + idle	io_client.py CPU §10.4 #5
	cron watchdog CLOSING

P1 —

	SimpleInternalCall +
	DP + Car.weight_state +
MOVE dispatch	app.py + executor.py §10.4 #6
	app.py [car_id=0 [falsy [tid §10.4 #3
	init

P2 —

cron stop listener	reload §10.4 #7
create_task	9+ §10.4 #8
WS reconnect	§10.4 #9
	test_executor.py 18 §10.4 #10
ActionQueue.get_nowait()	/reset §10.4 #1
floor_display	§10.4 #2

P3 —

-
-
- human_presence
- + +
- `/algo set <name>`



```
executor.py brake-before-stop 100ms asyncio.sleep — sleep /
```

```
PLC [ asyncio.sleep(0.1) [ wait_for(signal, timeout=...) ]
```

PEC \$7.6PLC



github.com/xmb505/Haku_EET



Star	
Fork	
Pull Request	

Fork



11.1 AI

AI AI

	DeepSeek

AI

11.2 AI

AI

WorkBuddy	MCP
Qoder	MCPQoder CN
TIA Portal MCP	github.com/Czarnak/tia-portal-mcp

MCPModel Context Protocol

TIA Portal MCPAIPLCAI

11.3

		7

-
- HMI
-